Degree-five Independence and a Nonzero Explicit Jacobian Class

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Abstract

For a field K and $k \neq 0$, four specified degree-five monomials are proved linearly independent in the actual quotient $K[x,y]/(5kx^4y, kx^5+6y^5)$. The proof uses coefficient functionals annihilating every element of the full ideal, with arbitrary polynomial multipliers. In characteristic zero, the previously checked section identity and exact coefficient certificates then imply that the actual quotient class of HW is nonzero at every root of the specified modulus. This is an algebraic milestone, not a geometric residue, Hodge or cycle-class theorem.

1 The actual quotient and the claim

Put $f_k = kx^5y + y^6$ and

$$I_k = (\partial_x f_k, \partial_y f_k) = (5kx^4y, kx^5+6y^5), \qquad Q_k = K[x,y]/I_k.$$

Brackets denote the quotient map. The functions `binarySextic`, `jacobianIdeal`, `Jacobian`, `qx` and `qy` in `Hodge/ExplicitJacobianQuotient.lean` are these actual objects, not an abstract interface with a postulated basis.

Theorem 1. *For every field K and $k \in K \setminus \{0\}$, the classes*

$$[y^5], \quad [xy^4], \quad [x^2y^3], \quad [x^3y^2]$$

are linearly independent over K in Q_k.

No nontriviality of Q_k, no basis hypothesis and no geometric realization are assumed. The theorem concerns four classes of degree five, not a basis of the entire quotient. It does not assert a formalized spanning or dimension theorem for a graded component either.

2 Coefficient functionals on the full ideal

For a polynomial p, let $c_{a,b}(p)$ be its coefficient of x^ay^b. Define

$$L_0(p) = kc_{0,5}(p) - 6c_{5,0}(p), \qquad L_i(p) = c_{i,5-i}(p) \quad (1 \le i \le 3).$$

The minus sign and the x^5 coefficient are essential: the raw coefficient of y^5 does not annihilate kx^5+6y^5 in characteristic zero.

Lemma 1. *For every commutative coefficient ring, every k and every $p \in I_k$, one has $L_0(p) = L_1(p) = L_2(p) = L_3(p) = 0$.*

Proof. Write $p = A(5kx^4y) + B(kx^5 + 6y^5)$, where A, B are arbitrary polynomials, with no degree restriction. Coefficient multiplication gives

$$c_{1,4}(p) = c_{2,3}(p) = c_{3,2}(p) = 0,$$
$$c_{0,5}(p) = 6c_{0,0}(B), \qquad c_{5,0}(p) = kc_{0,0}(B).$$

For example, the generator x^4y cannot contribute at any of the first three positions because its x exponent is already too large. It cannot contribute to either pure-power position. At the two pure powers, only the constant term of B contributes; every other multiplier monomial would increase an exponent. Thus $L_0(p) = 6kc_{0,0}(B) - 6kc_{0,0}(B) = 0$. This proves the claim for the full ideal, not just its listed generators or constant linear combinations thereof. □

In Lean, the proof uses `Ideal.mem_span_pair` and `MvPolynomial.coeff_mul_monomial'`. These handle arbitrary ideal multipliers exactly. The finite exponent checks do not postulate a homogeneous basis or a Groebner basis.

Proof of Theorem 1. Suppose $[\Phi(c)] = 0$, where

$$\Phi(c) = c_0y^5 + c_1xy^4 + c_2x^2y^3 + c_3x^3y^2.$$

Then $\Phi(c) \in I_k$. Lemma 1 gives $c_1 = c_2 = c_3 = 0$ and $kc_0 = 0$. Since K is a field and $k \neq 0$, also $c_0 = 0$. This is precisely linear independence in the actual quotient. □

3 Application to the actual explicit section

For this application let K be a characteristic-zero field equipped with its specified $\mathbb{Q}$-algebra structure, and suppose

$$P(k) = k^{24} + 66081312000k^{18} - 4811512860000k^{12} + 1167566400000k^6 + 583200000 = 0.$$

The nonzero constant term implies $k \neq 0$; this is a previously checked theorem, not an extra nontriviality assumption.

Use the actual coefficient functions b, c, d, e, g from `Hodge/ExplicitCurve.lean`, and set

$$H = x^2 + bxy + cy^2, \qquad G = x^3 + dx^2y + exy^2 + gy^3, \qquad W = H_xG_y - H_yG_x.$$

Their exact rational-polynomial definitions, together with every reduction and Bezout certificate, are included in the pinned proof sources. In particular, the computation is reproducible without numerical root approximation. Writing $W = w_3x^3 + w_2x^2y + w_1xy^2 + w_0y^3$, the checked formulas are

$$w_3 = 2d - 3b = k, \qquad w_2 = 4e - bd - 6c,$$
$$w_1 = 6g + be - 4cd, \qquad w_0 = 3bg - 2ce.$$

In the actual quotient, $[x^4y] = 0$ and $k[x^5] + 6[y^5] = 0$. Expanding HW and using these relations gives the already checked identity

$$[HW] = u_0[y^5] + u_1[xy^4] + u_2[x^2y^3] + u_3[x^3y^2],$$

where

$$u_0 = cw_0 - 6, \qquad u_1 = bw_0 + cw_1,$$
$$u_2 = w_0 + bw_1 + cw_2, \qquad u_3 = w_1 + bw_2 + ck.$$

The -6 is the contribution of $k[x^5]$, not an omitted or freely chosen term.

The earlier exact polynomial certificates prove $u_0(k) \neq 0$ whenever $P(k) = 0$. Their mechanism is $u_0 = QP + r$ and $Ar + BP = 1$ in $\mathbb{Q}[T]$, with the explicit polynomials recorded in `ExplicitResidueCertificates.lean`. Evaluating at a root gives $A(k)u_0(k) = 1$. This is an actual checked polynomial identity, not a numerical test or a new coefficient-nonvanishing hypothesis.

Theorem 2. *For every such field and every root k of P, the actual class $[HW] \in Q_k$ is nonzero.*

Proof. If $[HW] = 0$, Theorem 1 and the displayed actual quotient identity imply $u_0(k) = 0$, contrary to the exact certificate. In fact only this one coefficient certificate is needed here, although the earlier stage proves all four nonzero. □

This statement is nonvacuous: the checked modulus has degree 24 and has an actual complex root. The new existential theorem supplies such a root together with its nonzero $[HW]$, without assuming that the quotient is nontrivial.

4 Formalization, reproduction and boundary

Lean source and exact versions

The module is `Hodge/ExplicitJacobianIndependence.lean` in `https://github.com/BenFrohman/HODGE`, code commit `52325b02f966139c99015443ca5fba8a725ef3bb` (pull request #49), on top of `main` at `b13f57f`. Toolchain `leanprover/lean4:v4.22.0`; Mathlib tag `v4.22.0`, locked revision `79e94a093aff4a60fb1b1f92d9681e407124c2ca` (`lake-manifest.json`). The full source is in the repository; nothing here is a summary or certificate in place of source. The module is imported by `Hodge.lean`, so the default library build checks it, and the CI sorry scan covers it.

Lean hypotheses versus the assumptions of this note

Lean hypothesis	Assumption in this note
`[Field K]`, `(hk : k ≠ 0)`	Theorem 1: K a field, $k \neq 0$.
`jacobianIdealk` (Stage 3 definition)	$I_k = (5kx^4y,\ kx^5 + 6y^5)$, the full ideal.
`[CharZero K] [Algebra ℚ K]`	"characteristic zero". For a field of characteristic 0 a $\mathbb{Q}$-algebra structure exists and is unique, so this adds nothing.
`hm : k^24+66081312000*k^18-...=0`	$P(k) = 0$, the degree-24 modulus.
none (existential)	`exists_complex_nonzero_HW_class`: a complex root of P with $[HW] \neq 0$.

No hypothesis is equivalent to its conclusion, and none is uninhabited: $k = 1$ over $\mathbb{Q}$ satisfies Theorem 1, and the last row exhibits a root for the second theorem.

One-command rebuild and axiom output

From the repository root at the commit above:

```
lake exe cache get && lake build && python3 scripts/check_residue_axioms.py
```

GitHub Actions ran exactly this (workflow `certify.yml`, run `https://github.com/BenFrohman/HODGE/actions/runs/42922528702`, passed; library build and sorry check: run `https://github.com/BenFrohman/HODGE/actions/runs/42922528652`, passed). Its axiom output for this module, one line per theorem in source order:

```
'Hodge.ExplicitJacobianIndependence.monomial_exponent'
'Hodge.ExplicitJacobianIndependence.partial_x_monomial'
'Hodge.ExplicitJacobianIndependence.partial_y_monomial'
'Hodge.ExplicitJacobianIndependence.jacobian_ideal_coefficient_constraints'
'Hodge.ExplicitJacobianIndependence.representative_coefficient'
'Hodge.ExplicitJacobianIndependence.representative_x5_coefficient'
'Hodge.ExplicitJacobianIndependence.representative_mem_ideal_iff'
'Hodge.ExplicitJacobianIndependence.representative_class_formula'
'Hodge.ExplicitJacobianIndependence.representative_class_sum'
'Hodge.ExplicitJacobianIndependence.representative_class_eq_zero_iff'
'Hodge.ExplicitJacobianIndependence.degreeFiveClasses_linearIndependent'
'Hodge.ExplicitJacobianIndependence.section_HW_class_ne_zero'
'Hodge.ExplicitJacobianIndependence.exists_complex_nonzero_HW_class'
  each: depends on axioms: [propext, Classical.choice, Quot.sound]
PASS: 78 incremental theorem declarations; standard axioms only
```

The root audit in the same run printed `audited699declarations,699withstandardaxiomsonly`. No `sorry`, `admit`, `native\_decide` or new `axiom` occurs. A kernel check shows that the stated Lean propositions follow from the stated definitions; it does not by itself establish any geometric statement.

No geometric cycle-class map, residue-to-cohomology comparison, Hodge-type identification, descent argument, or geometric counterexample is proved by this milestone. In particular a nonzero class in this binary polynomial quotient must not be relabeled a nonalgebraic Hodge class. The remaining geometric bridges require their actual definitions and proofs.

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